BCS-074	Android Operating System	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To introduce Android platform and its architecture.
CO2	To learn activity creation and Android UI designing.
CO3	To be familiarized with Intent, Broadcast receivers and Internet services.
CO4	To integrate multimedia, camera and Location based services in Android Application.
CO5	To learn about publishing

Detailed Syllabus

MODULE-I

About Android , Smart phones future, **Preparing the environment-** Installing the SDK , Creating Android Emulator , Installing Eclipse , Installing Android Development Tools , Choosing which Android version to use **Android Architecture-** Android Stack, Android applications structure

MODULE 2

UI Architecture-Application context , Intents ,Activity life cycle , Supporting multiple screen sizes **User Interface Widgets**- Text controls , Button controls ,Toggle buttons , Images **Notifications and Toasts-** Parameters on Intents ,Pending intents ,Status bar notifications ,Toast notifications **Menus-** Localization, Options menu , Context menu **Dialogues-** Alert dialog , Custom dialog , Dialog as Activity

MODULE 3

Lists-Using string arrays, Creating lists, Custom lists **Location and Maps-** Google maps , Using GPS to find current location **Working of Data Storages-**Shared preferences ,Preferences activity ,Files access , SQLite database **Network Communication**-Web Services , HTTP Client , XML and JSON **Services-**Service lifecycle, Foreground service **Publishing the App**-Preparing for publishing ,Signing and preparing the graphics ,Publishing to the Android Market.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Describe Android platform, Architecture and features.
CO2	Design User Interface and develop activity for Android App.
CO3	Use Internet, Broadcast receivers and Internet services in Android App.
CO4	Use multimedia, camera and Location based services in Android App.
CO5	Understand about Publishing

Text Books:

1. Bill Philips & Brian Hardy, Android Programming: The Big Nerd Ranch Guide
2. Greg Nudelman, Android Design Patterns: Interaction Design Solutions for Developers
3. Ian G. Clifton, Android User Interface Design: Turning Ideas and Sketches into Beautifully Designed Apps
4. Ed Burnette, Hello, Android: Introducing Google's Mobile Development Platform (Pragmatic Programmers).